Financial Statement Analysis

Financial Statement Analysis

- Financial Analysis enables us to gauge the returns and risks, both, present and future and hence ensures in improving decision-making.
- Analysis of financial statements is like solving a puzzle. One should not rush to make expansive conclusions based on one or two clues / pieces of puzzle.

Financial Statement

- The balance sheet reveals the financial position of the business in terms of the assets owned and the different claims against these assets held by outsiders and owners.
- The profit and loss account summarizes the revenues and expenses of an accounting period. It gives the cost structure of the business and the relationship of the costs to the revenues.

Features of Financial Statement

1. Financial Statements are prepared at the end of the accounting period.
2. Financial Statements disclose both facts and opinions.
3. Financial statements are prepared on the going concern value.
4. Financial statements are recorded facts of financial transactions based on historical cost.
5. Financial statements are greatly affected by personal judgement of the accountants.

FSA Methods

- Common Size Analysis
- Index Based Analysis
- Ratio Analysis

Common size financial statement

- Away to analyze a firm is to look at its financial data more meticulously. Financial statements presented by representing each item as a percentage to the total amount of which it is a part, is referred to as a ‘common size financial statement’.

Common Size Balance Sheet

- A common size balance sheet is constructed by showing each item of asset as a percentage of the total assets, and similarly, each item of liability and owners’ equity is shown as a percentage of the total liabilities and owners’ equity.

Example

TOOLS & TOOLS LIMITED COMMON SIZE BALANCE SHEET As at December 31, 20X7				
	20X7		20X6	
ASSETS	**₹ million**	**%**	**₹ million**	**%**
Cash	19	5.76	11	4.07
Accounts receivable	32	9.70	20	7.41
Loans and advances	43	13.03	34	12.59
Inventory	121	36.67	99	36.67
Other current assets	17	5.15	26	9.63
Total current assets	**232**	**70.31**	**190**	**70.37**
Fixed assets(Net)	94	28.48	79	29.26
Other assets	4	1.21	1	0.37
Total assets	330	100.00	270	100
LIABILITIES & CAPITAL				
Current Liabilities:				
Acceptances	5	1.52	2	0.74
Accounts payable	27	8.18	19	7.04
Advance against sales	26	7.88	21	7.78
Other liabilities	9	2.73	8	2.96
Interest accrued but not due	3	0.90	2	0.74
Total current liabilities	70	21.21	52	19.26
Provisions:				
for taxation	26	7.88	21	7.77
for proposed dividends	2	0.61	2	0.74
for bonus	3	0.90	2	0.74
for others	4	1.21	3	1.11
Total provisions	35	10.61	28	10.37
Total current liabilities and provisions	105	31.82	80	29.63
Long-term liabilities				
Bank term loans	40	12.12	32	11.85
10.5% Debentures	26	7.88	26	9.63
Financial institutions	24	7.27	22	8.15
Total long-term liabilities	90	27.27	80	29.63
Total liabilities	195	59.08	160	59.26
Shareholders' equity				
Paid up capital	37	11.21	37	13.70
Retained earnings	98	29.70	73	27.04
Total shareholders' equity	135	40.91	110	40.74
Total Liabilities & Shareholders' Equity	330	100	270	100

Specific case- Assets and Liabilities

TOOLS & TOOLS LIMITED Structure of assets and liabilities		
	20X7 in %	**20X6 in %**
Total current assets	70.31	70.37
Fixed assets (Net)	28.48	29.26
Other assets	1.21	0.37
Total assets	100.00	100.00
Total current liabilities and provisions	31.82	29.63
Total long-term liabilities	27.27	29.63
Total liabilities	59.09	59.26
Total shareholders' equity	40.91	40.74
Total Liabilities & Shareholders' Equity	100	100

Working capital and Net Income

Working Capital Change as Percentage of Total Assets			
	20X7	**20X6**	**Change in WC**
Total current assets	70.31	70.37	–0.07
Total current liabilities and provisions	31.82	29.63	–2.19
Working capital	38.49	40.74	–2.25

TOOLS & TOOLS LIMITED Dis-aggregation of Change in Net Income	
	% Change in expenses and net income in relation to sales.
Decrease in cost of goods sold	0.67
Decrease in general expenses	0.71
Total decrease in cost increasing net income	**1.38**
Increase in selling expense	0.47
Increase in interest expense	0.03
Increase in income tax	0.46
Cost increases decreasing net income	**0.96**
Net Change in Net income	**0.42**

Common Size Profit and Loss Account

- The profit and loss account summarizes the revenues and expenses of an accounting period. We can prepare a commons size profit and loss account by showing the net sales as 100% and each of the components of expenses and profits as a percentage of the net sales.

Example

TOOLS & TOOLS LIMITED
COMMON SIZE PROFIT & LOSS ACCOUNT
For the year ended December 31, 20X7

	20X7		20X6	
	₹ million	%	₹ million	%
Sales	300	100	280	100
Cost of goods sold	148	49.33	140	50.00
Gross profit	152	50.67	140	50.00
Selling expenses	25	8.33	22	7.86
General expenses	60	20.00	58	20.71
Total operating expenses	85	28.33	80	28.57
Operating income	67	22.33	60	21.43
Interest expense	14	4.67	13	4.64
Net income before tax	53	17.67	47	16.79
Income tax	26	8.67	23	8.21
Profit after tax	27	9.00	24	8.57
Dividends	2	0.67	2	0.71
Profit retained	25	8.33	22	7.86

Index-based Balance Sheet

- In an indexed based analysis we represent the accounting information over multiple years as a percentage of amounts of an observant base year. It helps to compare accounts for two or more years to the corresponding items for a single company. The base year figures of each item are always indexed to 100, and the changes from the base year are determined.

Example

Table 7.1 Index Based Balance Sheet Colgate Palmolive (India) Limited

Balance Sheet Items				
Assets Side	**Year 4**	**Year 3**	**Year 2**	**Year 1**
Cash & Bank Balances	123.4	89.1	113.7	100.0
Accounts Receivable	831.3	893.3	770.8	100.0
Loans and Advances	79.5	88.7	66.7	100.0
Inventories	167.6	196.9	139.0	100.0
Current Assets	134.2	122.5	119.5	100.0
Fixed Assets	151.1	127.9	104.0	100.0
Other Assets	337.0	211.0	166.3	100.0
Total Assets	**148.0**	**128.0**	**117.1**	**100.0**
Liabilities Side	**Year 4**	**Year 3**	**Year 2**	**Year 1**
Accounts Payables	124.8	98.7	101.2	100.0
Other Current Liabilities and Provisions	176.8	165.3	136.7	100.0
Current Liabilities and Provisions	141.5	120.2	112.7	100.0
Long-term Liabilities	779.2	671.9	606.0	100.0
Net Worth	150.1	133.5	117.8	100.0
Total Liabilities	**148.0**	**128.0**	**117.1**	**100.0**

Balance Sheet Item	Year 4	Year 3	Year 2	Year 1
Cash and Bank Balances	4,288 millions	3,098 millions	3,951 millions	3,476 millions

$$\text{Common Base Cash in Year 4} = \frac{\text{Cash in Year } 4 \times 100}{\text{Cash in Year 1}} = \frac{4288 \times 100}{3476} = 123.3$$

Connections between Income Statement and Balance Sheet

Income Statement of Colgate-Palmolive (India) Ltd. for Year 4	₹ million
Total Sales	**31,638**
Add : Other Income	499
Less: Cost of Goods Sold	12,501
Gross Profit	**19,636**
Less : Operating/Other Expenses	12,556
EBITDA	**7,080**
Less : Interest	13
Less : Depreciation & Amortization	437
Profit before Tax	**6,630**
Less : Tax	1,663
Net Income	**4,968**
Dividend to Shareholders	**3,808**

Balance Sheet of Colgate-Palmolive (India) Limited for Year 4	
Assets Side	**₹ million**
Cash & Bank Balances	4,288
Accounts Receivable	812
Loans and Advances	977
Inventories	1,853
Current Assets	**7,930**
Fixed Assets	**3,826**
Other Assets (Investments, Goodwill, etc.)	**1,311**
Total Assets	**13,068**
Liabilities Side	
Accounts Payables	4,666
Other Current Liabilities & Provisions	3,148
Total Current Liabilities & Provisions	**7,814**
Long-term Liabilities	**357**
Net Worth	**4,896**
Total Liabilities	**13,068**

Index Based Profit and Loss Statement

	Colgate Palmolive (India) Limited			
Profit and Loss Statement Items	**Year 4**	**Year 3**	**Year 2**	**Year 1**
Total Sales	**161.2**	**137.2**	**113.2**	**100.0**
Add : Other Income	**50.7**	**51.5**	**108.5**	**100.0**
Less: Cost of Goods Sold	160.9	135.2	112.2	100.0
Gross Profit	152.9	131.9	113.3	100.0
Less : Operating/Other Expenses	165.2	140	113.3	100.0
EBITDA	**135.1**	**120.1**	**106.1**	**100.0**
Less : Interest	86.4	100.6	106.4	100.0
Less : Depreciation	116.3	104.6	91.2	100.0
Profit before Tax	136.8	121.4	107.3	100.0
Less : Tax	270.2	230.7	190.7	100.0
Net Income	**117.4**	**105.5**	**95.1**	**100.0**
Dividend to Shareholders	140.0	125.0	110.0	100.0

Problems with Financial Statement Analysis

Comparability between periods. The company preparing the financial statements may have changed the accounts in which it stores financial information, so that results may differ from period to period. For example, an expense may appear in the cost of goods sold in one period, and in administrative expenses in another period.

Comparability between companies. An analyst frequently compares the financial ratios of different companies in order to see how they match up against each other. However, each company may aggregate financial information differently, so that the results of their ratios are not really comparable. This can lead an analyst to draw incorrect conclusions about the results of a company in comparison to its competitors.

Operational information. Financial analysis only reviews a company's financial information, not its operational information, so you cannot see a variety of key indicators of future performance, such as the size of the order backlog, or changes in warranty claims. Thus, financial analysis only presents part of the total picture.

3. Financial Ratio Analysis

PROFITABILITY	Margin on sales	Gross Profit Margin
		Operating Profit Margin
		Earnings before Interest & Tax
		Profit before Tax
		Net Profit Margin (i.e., Profit after tax)
	Return on investment	Operating Profit to Operating Assets
		Net Income to Total Assets
		Return on Equity
	Efficiency	Total Asset Turnover
		Operating Asset Turnover
		Working Capital Turnover
		Shareholder Equity Turnover
	Return per share	Earnings per Share
		Earnings to price
		Dividends per Share

SOLVENCY	Short-term	Net Working Capital
		Current Ratio
		Quick Ratio
		Accounts Receivable Turnover
		Collection Period
		Inventory Turnover
		Conversion Period
	Long-term	Total Debt to Total Capital
		Long Term Debt to Total Capital
		Long Term Debt to Fixed Assets
		Interest Cover
		Times Fixed Charges Covered
		Gearing
		Equity Multiplier

Formula and interpretation

(i) **Gross Profit Margin:** The gross profit margin is the **surplus available out of sales revenues, after subtracting the cost of goods sold.**

$$\text{Gross Profit} = \text{Sales} - \text{COGS}$$

$$\text{Gross Profit Margin}\,(\mathbf{GPM}) = \frac{\text{Gross Profit} \times 100}{\text{Sales}}$$

(ii) Operating profit margin: The operating margin is a reflection of the operations of the company, and hence, is considered as a ***reflection of the performance of the management.*** Operating expenses are mostly fixed costs of operations, such as general, marketing, and administrative expenses. The level or operating profits will be determined by the company's ability to maintain a sufficient volume of sales and gross margin. This margin is frequently used as a basis of ***comparison across companies***, since it is not influenced by different financing decisions.

$$\text{Operating Profit}^{8} = \text{PBDIT} - \text{Depreciation}$$

$$\text{Operating Profit Margin}\,(\mathbf{OPM}) = \frac{\text{Operating Profit} \times 100}{\text{Sales}}$$

[8] Operating profit does not include the Other Income.

Ratios

(iii) Earnings before interest and taxes: Any non-operating surplus or deficit is adjusted to the operating profit margin, to obtain the earnings before interest and taxes.

(iv) Profit before taxes: This is the amount of surplus obtained after meeting the interest expense. This is influenced to a great extent, by the financing decisions of the firm. This shows the percentage of sales revenue that is available before taxation.

(v) Profit after tax: The profit after tax or net income is the overall surplus available out of the sales. This is also the amount available to the shareholders. This is influenced by three major factors, namely, ***operating efficiency, financing efficiency and taxation.*** This amount, which the company is able to make, is evaluated in relation to the amount of assets invested in the operations and the amount invested by shareholders. As a percentage of the sales, it is often known as the *Net Profit Margin* and is also used to ***compare margins of various players in the same industry.***

$$\text{Net Profit Margin}\,\textbf{(NPM)} = \frac{\text{Net Profit} \times 100}{\text{Sales}}$$

EBIT = Revenue – COGS – Operating Expenses

Return on Investment

$$\text{Return on Operating Assets}\,\textbf{(ROA)} = \frac{\text{Operating Profit} \times 100}{\text{Average Operating Assets}}$$

Return on Operating Assets (ROA)	20X7	20X6
Current assets (₹ million)	232	190
Fixed assets (₹ million)	94	79
Total operating assets (₹ million)	326	269
Operating Profit Before Interest and Taxes (OPBIT) (₹ million)	67	60
Return on Operating Assets (%)	20.55	22.30
ROA based on average operating assets (326 + 269)/2(%)	22.52	

Return on Total Assets

$$\text{Return on Total Assets}\,\textbf{(ROTA)} = \frac{\text{Net Profit} \times 100}{\text{Average Total Assets}}$$

Return on Total Assets (ROTA)	20X7	20X6
Total Assets (₹ million)	330	270
Profit After Taxes (PAT)	27	24
Return on Total Assets	8.18%	8.89%
ROTA based on average total assets (330 + 270)/2	9.00%	

Return on Equity

Return on Equity: This measures the net income (or PAT) as a percentage of the shareholders' investment. Net income after tax is the amount available to the shareholders for compensating their investment and the risk being carried by them.

$$\text{Return on Equity } \mathbf{(ROE)} = \frac{\text{Net Income} \times 100}{\text{Average Net Worth}}$$

Return on Equity (ROE)	**20X7**	**20X6**
Total Equity (₹ million)	135	110
Profit After Taxes (PAT)	27	24
Return on Equity (%)	20.00	21.82
ROE based on average total assets (135 + 110)/2(%)	22.04	

Earnings per Share

Earnings per Share: This is one of the most commonly used measures of expressing corporate earnings. This measure is computed by dividing the net income by the number of equity or ordinary shares outstanding.

$$\text{Earnings Per Share}\,(\textbf{EPS}) = \frac{\text{Net Income (PAT)}}{\text{Number of Equity Shares } (n)}$$

Earnings per Share (EPS)	20X7	20X6
Profit After Taxes (PAT) (₹ million)	27	24
Number of ordinary shares (Million)	3.7	3.7
Earnings per Share (EPS) (₹)	7.30	6.49

Price Earnings (PE) Ratio

$$\text{Earnings Price Ratio} = \frac{\text{Earnings per share}}{\text{Market price per share}}$$

$$\text{Price-Earnings Ratio} = \frac{\text{Market Price Per Share}}{\text{EPS}} = \frac{\text{Market Capitalization}}{\text{Net Income}}$$

Earnings Price Ratios (E/P)	20X7	20X6
Earnings per Share (EPS) (₹)	7.30	6.49
Market price per share (₹)	30	28
Earnings/Price Ratio (%)	24.3	23.18
Price Earnings Ratio	4.1	4.31

Dividend per share

$$\text{Dividend Per Share } \mathbf{(DPS)} = \frac{\text{Dividend}}{\text{Number of Equity Shares } (n)}$$

Dividend per Share (DPS) (₹)	20X7	20X6
Dividend (₹ Million)	2	2
Number of ordinary shares (Million)	3.7	3.7
Dividend per Share (₹)	0.54	0.54

can be more than 100% → lead to erosion of net worth

Dividend Payout Ratio

- Dividend Payout Ratio: Net Income of a company is either distributed or retained. Dividend payout ratio is the portion of the net income that gets distributed with the equity shareholders as dividend.

$$\text{Dividend Payout Ratio} = \frac{\text{Dividend} \times 100}{\text{Net Income}} = \frac{n \times \text{Dividend Per Share} \times 100}{n \times \text{Earnings Per Share}}$$

Hence, this also can be written as,

$$\text{Dividend Payout Ratio} = \frac{\text{Dividend Per Share}}{\text{Earnings Per Share}}$$

Efficiency in Asset Utilization

$$\text{Return on Total Assets (ROTA)} = \frac{\text{Net Profit}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Average Total Assets}}$$

Tools & Tools Limited: Computing Profitability Component

Asset Utilization and Returns	20X7	20X6
Total Assets (₹ million)	330	270
Profit After Taxes (PAT) (₹ million)	27	24
Sales(₹ million)	300	280
Net Income/ Sales (%)	9.00	8.57
Sales / Total Assets (also known as Asset Utilization Ratio)	0.91	1.04
Return on Total Assets (Net Income /Total Assets) (%)	8.18	8.89
Sales / Average Total Assets	1.00	

Turnover Ratio

$$\text{Working Capital Turnover} = \frac{\text{Sales}}{\text{Average Working Capital}}$$

$$\text{Shareholders' Equity Turnover} = \frac{\text{Sales}}{\text{Average Shareholders' Equity}}$$

Solvency Ratio

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilites}}$$

Quick ratio: Quick ratio or acid test ratio is usually computed by taking assets, which are quick to be converted into cash and divide them by the current liabilities.

Quick ratio	20X7	20X6
Current assets (₹ million)	232	190
Inventory (₹ million)	121	99
Quick Assets (₹ million)	111	91
Current Liabilities (₹ million)	105	80
Net Working Capital (₹ million)	127	110
Quick ratio: Quick Assets/Current Liabilities	1.06	1.14

Exercise

- Vinyl Chemicals (India) Limited is a part of the Vinyl group of companies. It is well known for its Fevicol brand name and its flagship company, Pidilite Industries Limited. The principal activity of the company is to manufacture VINYL Acetate Monomer. The company is also engaged in trading of chemicals and the production of processed air.
- Following are the balance sheets and the profit and loss accounts of VINYL Chemicals for two years. Please give your comments on the solvency and profitability position of the company, for the financial year (FY) 20X6 (compared to FY 20X5).

Financial Details

VINYL Chemicals Limited Balance Sheet on 31 March

PARTICULARS	All figures in ₹ million	
Assets Side	**FY 20X6 (12m)**	**FY 20X5 (12m)**
Cash & Bank Balances	1.01	1.00
Accounts Receivable	16.75	13.01
Loans and Advances	3.76	3.44
Inventories	11.53	11.98
Current Assets	33.05	29.43
Fixed Assets	37.22	32.43
Investments	3.42	4.41
Total Assets	**73.69**	**66.27**
Liabilities Side		
Current Liabilities & Provisions	8.18	8.63
Long-term Liabilities	12.32	8.08
Net Worth (Total Shareholders' funds)	53.18	49.55
No. of Equity Shares of ₹10 par value	1.83	1.83
Total Liabilities	**73.68**	**66.26**

VINYL CHEMICALS (INDIA) LIMITED
Profit & Loss Account on 31 March

(All figures in ₹ million)

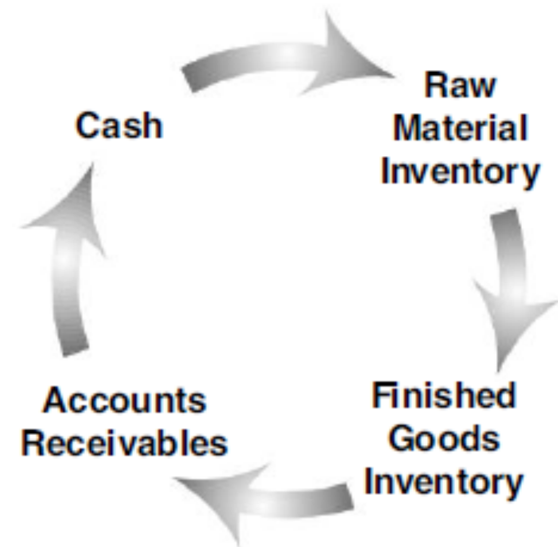
PARTICULARS	FY 20X6 (12m)	FY 20X5 (12m)
Total Sales	92.46	88.18
Other Income	3.18	1.11
Operational Expenditure	84.64	87.62
Gross Profit	10.62	1.03
Interest	0.82	0.58
Depreciation	2.53	2.42
Profit Before Tax	7.27	–1.97
Tax	1.62	0.00
Profit After Tax	5.65	–1.97
Dividend to Shareholders	1.83	0.00

Solution

VINYL Chemicals: Financial Ratios			FY 20X6	FY 20X5
Profitability	Margin on sales	Gross Profit Margin (in %)	11.49	1.17
		Profit before tax (in %)	7.86	–2.23
	Return on Investment	Net Income to Total Assets (in %)	7.67	–2.97
		Return on Equity (in %)	10.62	–3.98
	Efficiency	Total Asset Turnover	1.32	
		Working Capital Turnover	4.05	
	Return per share	Earnings per share (in ₹)	3.09	–1.08
Solvency	Short term	Net Working Capital (in ₹ millions)	24.87	20.80
		Current Ratio	4.04	3.41
		Quick Ratio	2.63	2.02
		Accounts Receivable Turnover	6.21	
		Collection Period (in days)	58.74	
		Inventory Turnover	7.20	
		Conversion Period (in days)	50.69	
	Long term	Total-debt to Total Capital	0.39	0.34
		Long-term Debt to Total Capital	0.23	0.16
		Interest Cover	9.87	–2.39

operating cycle → duration of time taken by unit of cash to circulate through business operations & return back as cash

Short-Term Solvency: Operating Cycle Perspective



$$\text{Accounts Receivable Turnover} = \frac{\text{Sales}}{\text{Average Accounts Receivable}}$$

$$\text{Average Collection Period (ACP)} = \frac{\text{Number of Days in the Accounting Period}}{\text{Accounts Receivable Turnover}}$$

$$\text{Average Collection Period} = \frac{\text{Accounts Receivable}}{\text{Sales}} \times \text{Days in the Period}$$

$$\text{Inventory Turnover} = \frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$$

$$\text{Inventory Holding Period} = \frac{\text{Average Inventory}}{\text{Cost of Goods Sold}} \times 365$$

↳ Diff

Continue.........

Diff

APP & Inventory

↳ COGS / Accounts payable

$$\text{Average Payables Period (APP)} = \frac{\text{Average Accounts Payable}}{\text{Cost of Goods Sold per day}}$$

$$\text{Accounts Payable Turnover} = \frac{\text{Cost of Goods Sold}}{\text{Average Accounts Payable}}$$

$$\text{Average Payables Period (APP)} = \frac{\text{Number of Days in the Accounting Period}}{\text{Accounts Payable Turnover}}$$

Operating Cycle and Cash Conversion Period

A I O

C O P

Operating Cycle = Inventory Coversion Period + Average Collection Period

This can also be written as:

Operating Cycles (in days) = ICP (in days) + ACP (in days)

$$\text{Operating Cycles Per Year} = \frac{365}{(\text{ICP} + \text{ACP})}$$

Cash Conversion Period (CCP) = Operating Cycle – Average Payables Period (APP)

- *Time taken to convert "cash → inventory → sales → accounts receivable" i.e., Inventory Conversion Period (ICP) does capture this period*
- *Time taken to collect accounts receivables "accounts receivable → cash" i.e., Average Collection Period (ACP) does capture this period*

CCP → +ve → greater than more need for liquidity

CCP → −ve → greater scope to use funds somewhere else

Long-term Solvency

$$\text{Total Debt to Equity Ratio} = \frac{\text{Current Liabilities} + \text{Long-term Liabilities}}{\text{Net Worth}}$$

$$\text{Long-term Debt to Equity Ratio} = \frac{\text{Long-term Liabilities}}{\text{Net Worth}}$$

$$\text{Long-term Debt to Fixed Asset} = \frac{\text{Long-term Debt}}{\text{Net Fixed Assets}}$$

Equity Multiplier

$$\boxed{\text{Equity Multiplier} = \frac{\text{Total Assets}}{\text{Net Worth}}}$$

$$\text{Equity Multiplier} = \frac{\text{Total Assets}}{\text{Net Worth}} = 1 + (\text{Total Debt to Equity Ratio})$$

$$\frac{\text{Total Assets}}{\text{Net Worth}} = \frac{\text{Total Debt} + \text{Net Worth}}{\text{Net Worth}}$$

$$\text{Equity multiplier} = \frac{\text{Total Debt}}{\text{Net Worth}}$$

$$= 1 + (\text{Total Debt to Equity Ratio})$$

	20X7	20X6
Total Debt (₹ million)	195	160
Shareholders Equity (₹ million)	135	110
Total Assets	330	270
Equity Multiplier (total assets/owners equity)	2.44	2.45
Return on Total Assets (%)	8.18	8.89
Return on Equity (ROTA × Equity Multiplier) (%)	20	21.8

DuPont Formula

- A DuPont analysis is an assessment of a company's Return on Equity (ROE). Using the DuPont formula, an analyst DuPont-Equation- determines a company's financial strengths and weaknesses. According to the DuPont analysis, ROE is affected by three elements:

$$\text{ROE} = \text{NPM} \times \text{Asset Turnover} \times \text{Equity Multiplier}$$

- Operating efficiency, which is measured by Net profit margin (NPM)
- Asset management, which is measured by total Asset Turnover
- Financial leverage, which is measured by the Equity Multiplier

$$\text{ROE} = \frac{\text{Net Income}}{\text{Shareholder Equity}}$$

$$\text{ROE} = \frac{\text{Net Income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Shareholders' Equity}}$$

$$\text{ROE} = \frac{\text{Net Income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Assets}} \times \frac{\text{Assets}}{\text{Shareholders' Equity}}$$

$$\text{ROE} = \text{NPM} \times \text{Asset Turnover} \times \text{Equity Multiplier}$$

ROE = Net Profit Margin (Profit/Sales) x Return on Assets (Sales/Assets) x Financial Leverage (Assets/Equity)

Problem set 1

Ralite Company had net income for the year of \$20 million. It had 2 million shares of common stock outstanding, with a year-end market price of \$82 a share. Dividends during the year were \$5.74 a share.

Required:
Calculate the following ratios: (*a*) price/earnings ratio, (*b*) dividend yield, and (*c*) dividend payout.

Solution

$$\text{Price/earnings ratio} = \frac{\$82}{(\$20{,}000{,}0\ 00/2{,}000{,}0\ 00\ \text{shares})}$$

$$= \frac{\$82}{\$10}$$

$$= 8.2 \text{ times}$$

$$\text{Dividend yield} = \frac{\$5.74}{82}$$

$$= .07\ (7 \text{ percent})$$

$$\text{Dividend payout} = \frac{(\$5.74 \text{ x } 2{,}000{,}000\ \text{shares})}{\$20{,}000{,}00\ 0}$$

$$= \frac{\$11{,}480{,}000}{\$20{,}000{,}000}$$

$$= 57.4 \text{ percent}$$

Problem set 2

You are given the following data on two companies, M and N (figures are millions):

	M	N
Sales	$1,080	$1,215
Net income	54	122
Investment	180	405

Required:

a. Which company has the higher profit margin?
b. Which company has the higher investment turnover?
c. Based solely on the data given, in which firm would you prefer to invest?

Hint: investment turnover is ratio of sales to total investment

Solution

a. Profit Margins

$$\frac{\text{M} - \text{Net income}}{\text{M} - \text{Sales}} = \frac{54}{1{,}080} = .05\ (5\ \text{percent})$$

$$\frac{\text{N} - \text{Net income}}{\text{N} - \text{Sales}} = \frac{122}{1{,}215} = .10\ (10\ \text{percent})$$

N has the higher profit margin.

b. Investment Turnover

$$\frac{\text{M} - \text{Sales}}{\text{M} - \text{Investment}} = \frac{1{,}080}{180} = 6\text{x}$$

$$\frac{\text{N} - \text{Sales}}{\text{N} - \text{Investment}} = \frac{1{,}215}{405} = 3\text{x}$$

M has the higher investment turnover.

c. Return on Investment

$$\frac{\text{M} - \text{Net income}}{\text{M} - \text{Sales}} \times \frac{\text{M-Sales}}{\text{M-Investment}} = .05 \text{ x } 6 = .30\ (30\ \text{percent})$$

$$\frac{\text{N} - \text{Net income}}{\text{N} - \text{Sales}} \times \frac{\text{N} - \text{Sales}}{\text{N} - \text{Investment}} = .10 \text{ x } 3 = .30\ (30\ \text{percent})$$

Both firms have similar returns on investments. Based on this investment criterion, the investments are equally attractive.